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Pace of ecology drives the tempo of visual perception across the animal kingdom

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The way an animal perceives its environment is fundamentally shaped by its ecology and evolution. The tempo of perception, the rate at which events can be perceived, varies widely across the animal kingdom and ranges from the rapid motion detection abilities of dragonflies to the sluggish perceptions of giant isopods. Autrum's hypothesis predicts that a species' pace of ecology drives this variation, but little is known about this relationship beyond taxon-specific cases. Here we use phylogenetic comparative methods to test the link between the tempo of visual perception and pace of ecology for 237 species, ranging from jellyfish to vertebrates. We show that higher temporal resolution is found in species with fast-paced ecologies associated with the ability to fly and with pursuit predation. We also find that the tempo of perception in ambush predators is mediated by environment context, which is probably related to the ability to act on fine-scale information. Our results highlight how ecology and the environment can shape the tempo of a species' perceptual experience.

For most animal species, visual perception is a crucial facilitator of environmental interactions, determining the ability to navigate, manoeuvre, forage and reproduce. However, sensory processing is costly, and natural selection is expected to select for efficient visual systems that detect and process only information that is relevant for the species' respective ecology[1]. As a result, the visual perception capabilities of different animal species are as diverse as the ecological niches they occupy. This diversity is apparent in many attributes of vision, such as spatial acuity, colour discrimination, depth perception and field of view, all of which vary widely among species[1,2]. This range in sensory attributes is in turn likely to create a diversity of sensory worlds, or Umwelts, in which organisms have differing experiences, even when they occupy the same habitat[3]. One such attribute that may profoundly affect a species' Umwelt is the tempo with which its visual processing occurs.

The tempo of visual processing arises from the limits of summating quanta of photon energy over an integration period, which requires a threshold to be reached before the information is processed and perceived by the individual[4,5]. The maximum rate at which this processing occurs can be generalized and quantified for any species, using the critical flicker fusion (CFF) threshold. CFF is defined as the maximum rate at which a flickering light can no longer be perceived as such and instead appears constant[6,7]. This measure of the tempo of perception has highlighted the extreme range of temporal abilities across visual systems in the animal kingdom. For example, CFF values of over 200 Hz have been measured for several invertebrates such as butterflies, blow flies and dragonflies[8,9], whereas species such as the crown-of-thorns starfish have maximum CFF values of less than 1 Hz (ref. 10). Although some of this variation is known to be linked to environmental conditions such as the need for longer summation times in low light conditions[11,12], much of this variation is yet to be explained. It is likely that CFF is shaped not solely by the environment but rather by a species' overall ecological profile.

One of the first formalized attempts to link ecology to the tempo of perception was Autrum's hypothesis, which proposed that the response dynamics of the retina should match a species habitat and lifestyle[13]. Several small-scale case studies have since supported this hypothesis. For example, high CFF thresholds have been typically found in fast-moving species such as pied fly catchers[14] and raptors[15].

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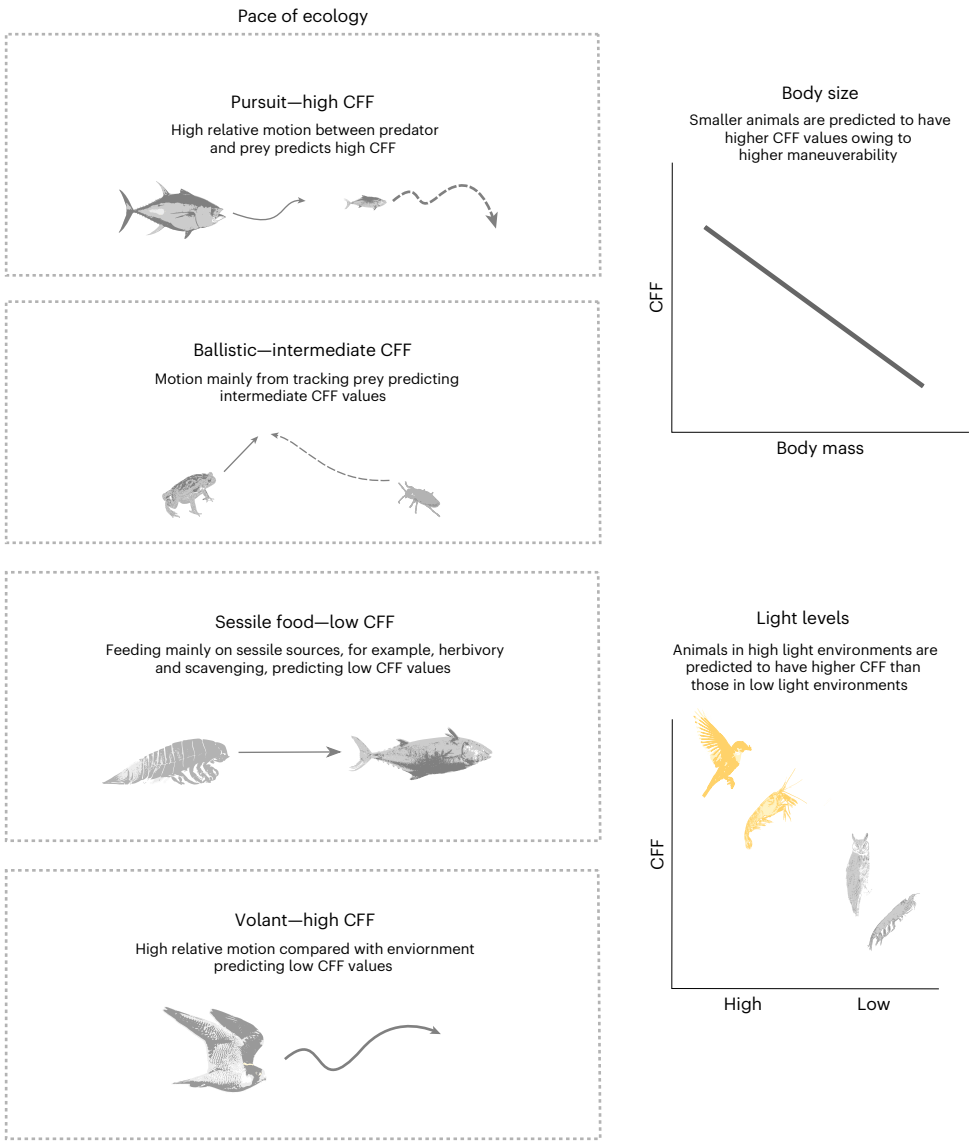


Fig. 1 | Predictions between species' pace of life and temporal perception. Species with fast ecologies such as volant species are expected to have a high tempo of visual perception as measured using CFF. To capture the pace of trophic ecologies, we split them into pursuit predation, ballistic predation and sessile-food foraging. Pursuit predation is defined as a continuous interaction between a predator and prey involving high levels of relative motions such as that found in tuna. Ballistic predation is defined by one-off interactions, typically involving a single attempt to capture prey with no prolonged pursuit of prey. Sessile-food foraging is defined as feeding on food sources which are stationary or passively moving through a forager's habitat (such as plankton) and includes strategies such as scavenging and grazing. We also predict that species associated with high-light environments, such as diurnal- or shallow-water species and smaller species have higher CFF values.

Furthermore, the highest CFF values within such fast-moving groups have been found in predators that pursue highly manoeuvrable prey[14,15]. Conversely, the lowest maximum CFF values are typically recorded in species with slow-paced ecologies, such as the crown-of-thorns starfish[10] and deep-sea isopods[16]. However, although there are several studies that support Autrum's[13] hypothesis within specific taxonomic groups, it has yet to be explored using larger scale phylogenetic comparative analysis that can test for the generality of such patterns.

Here, we categorize a species' ecological pace on the basis of its general and trophic ecology (Fig. 1), with a fast-paced ecology being characterized by qualities such as high manoeuvrability and a high-speed foraging strategy. In particular, we hypothesize that higher CFF values will be associated with volancy (flight) and smaller size owing to increased manoeuvrability within the environment[17]. We also hypothesize that the level of relative motion associated with a species trophic ecology will be associated with its CFF, with the lowest values associated with species that are not required to track movement, such as scavengers[16], and the highest CFF values associated with pursuit predators, such as pied fly catchers[14].

Results

Main results

We find that faster ecological modes of life, both in terms of foraging mode and volancy, were associated with higher CFF values (Tables 1–3, and Figs. 2 and 3a). Across all models, pursuit predators were associated with higher CFF values, with CFF values 1.55 times higher in pursuit predators when compared with sessile-food foraging species in the main model ($\beta = 0.19$, lower 95% confidence interval (CI) 0.02, higher 95% CI 0.35, back-transforming from the $\log_{10}$ scale gives $10^{0.19} = 1.55$; Table 1). Although pursuit predators were also associated with higher CFF values in both the terrestrial and aquatic models, these differences were not significant, probably owing to the small sample sizes

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Table 1 | Main model testing the role of log_{10} body mass (g), light levels, pace of ecology and habitat type on log_{10} CFF

Main model	Estimate (β)	Lower CI	Upper CI
Fixed effects			
Intercept	1.22	0.87	1.54
Log_{10} body mass	−0.01	−0.05	0.02
Light level low	**−0.31**	**−0.40**	**−0.22**
Log_{10} body mass: light level low	−0.01	−0.04	0.03
Pace of ecology ballistic predator	**0.12**	**0.02**	**0.21**
Pursuit predator	**0.19**	**0.02**	**0.35**
Habitat nektonic	0.19	−0.05	0.46
Demersal	0.16	−0.08	0.43
Volant	**0.33**	**0.13**	**0.55**
Method ERG	**0.07**	**0.00**	**0.14**
Random effects			
Phylogeny	0.11	0.07	0.21
Within species	0.03	0.01	0.04
Residuals	0.01	0.01	0.02

Significant values, highlighted in bold, are deemed to be those with 95% of the posterior estimate above or below zero. The model has 280 CFF measures for 237 species. The modes (β) and 95% CI of the posterior distributions are given for all fixed and random terms in a model with log_{10} of CFF as the response variable. Fixed terms include the continuous variable body mass (g) and the categorical terms light level (high, low), pace of ecology (sessile-food foraging, ballistic predator, pursuit predator), habitat (terrestrial, nektonic, demersal, volant) and CFF measurement type (behavioural, ERG). The random terms associated with phylogenetic relatedness (phylogeny), intraspecific variation (within species) and residual variation (residual) are also presented. The h^2 for this model is 0.79 with a lower and upper 95% CI of 0.63 and 0.88.

Table 2 | Model testing the role of log_{10} body mass (g), light levels, pace of ecology and habitat type on log_{10} CFF for terrestrial and volant species

Main model	Estimate (β)	Lower CI	Upper CI
Fixed effects			
Intercept	1.35	0.94	1.74
Log_{10} body mass	0.01	-0.05	0.06
Light level low	**−0.18**	**−0.29**	**−0.07**
Log_{10} body mass: light level low	−0.03	−0.08	0.03
Pace of ecology ballistic predator	−0.03	−0.25	0.18
Pursuit predator	0.14	−0.04	0.32
Habitat volant	**0.28**	**0.08**	**0.50**
Method ERG	**0.08**	**0.00**	**0.17**
Random effects			
Phylogeny	0.15	0.06	0.28
Within species	0.01	0.00	0.03
Residuals	0.02	0.01	0.03

Significant values, highlighted in bold, are deemed to be those with 95% of the posterior estimate above or below zero. The model has 143 CFF measures for 116 species. The modes (β) and 95% CI of the posterior distributions are given for all fixed and random terms in a model with log_{10} of CFF as the response variable. Fixed terms include the continuous variable body mass (g) and the categorical terms light level (high, low), pace of ecology (sessile-food foraging, ballistic predator, pursuit predator), habitat (terrestrial, volant) and CFF measurement type (behavioural, ERG). The random terms associated with phylogenetic relatedness (phylogeny), intraspecific variation (within species) and residual variation (residual) are also presented. The h^2 for this model is 0.86 with a lower and upper 95% CI of 0.69 and 0.94.

for pursuit species in these models ($n = 10$ in the terrestrial-only model, $n = 4$ in the aquatic-only model).

Similar to pursuit predators, species with volant lifestyles were also found to have significantly higher CFF values compared with non-volant species, with CFF values approximately twice as high in volant species in both the main ($\beta = 0.33$, lower 95% CI 0.13, higher 95% CI 0.55, Table 1) and terrestrial models ($\beta = 0.28$, lower 95% CI 0.08, higher 95% CI 0.50; Tables 1 and 2, and Fig. 2). Although nektonic species (species primarily associated with actively swimming in the water column) were associated with slightly higher CFF when compared with demersal species (species primarily associated with the bottom of their aquatic habitat), these differences were not significant (Table 3).

Predation strategy

Unlike pursuit predation, the effects we found associated with ballistic predators were mediated by the environment. Although in the overall model ballistic predators were associated with a significant increase of 1.32 in CFF compared with sessile-food foraging species ($\beta = 0.12$, lower 95% CI 0.02, higher 95% CI 0.21; Table 1), this association was found to have a negative, albeit non-significant, effect in the terrestrial-only model (Table 2). However, we found a significant positive increase for ballistic species of 1.38 times that of sessile-food foraging species in the aquatic-only model ($\beta = 0.14$, lower 95% CI 0.04, higher 95% CI 0.25; Table 3 and Fig. 3b) highlighting that the positive effect found in the main model is likely to be mainly driven by the association with marine but not terrestrial species.

Light level and body mass

As expected, CFF was associated with light levels across all models, with species in low light environments having significantly lower CFF values compared with species in high light environments across all models (Tables 1–3 and Figs. 1–3). For the effects of body mass, although we found that larger body masses were negatively associated with CFF values in all models, this effect was only significant in the aquatic model with CFF values reduced by approximately half over the five orders of magnitude range of body mass in aquatic species (Table 3). Interestingly, the positive interaction term between body mass and light environment indicates that this negative association between CFF and body mass is only found in high light environments (Table 3). Conversely, this interaction term results in a positive association with CFF in low light levels, although this overall effect size of 0.04 is lower in comparison with the negative effect size in high light environments (Table 3).

In all models, we found that CFF values measured using electroretinogram (ERG) methods were slightly, but significantly, higher in the main and terrestrial models, when compared with behavioural measurements (Tables 1 and 2). Finally, we found a strong phylogenetic effect in all three of the models with the h^2 value, which is comparable to Pagel's λ, ranging between 0.79 and 0.93, indicating that the level of species' relatedness has a significant effect on the amount of variation in temporal resolution in our dataset (Tables 1–3). In the supplementary models where we removed *Melanophila acuminata*, we find comparable results to the main models (Supplementary Section 1).

Discussion

Our results show that Autrum's hypothesis, which is that the tempo of visual perception is driven by the pace of a species' ecology[13], is supported at scales that span the animal kingdom. Although previous studies have provided support through exploring the tempo of perception of species with specifically fast[14] or slow ecologies[10] or through comparisons within taxonomic or ecological setting[16], we show for the first time that this support extends to the largest macroecological scales using phylogenetic comparative methods. By comparing species from disparate phyla, including Cnidaria, Annelida, Arthropoda,

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Table 3 | Model testing the role of log_{10} body mass (g), light levels, pace of ecology and Habitat type on log_{10} CFF for aquatic species

Main model	Estimate (β)	Lower CI	Upper CI
Fixed effects			
Intercept	1.37	0.95	1.79
Log_{10} body mass	**−0.05**	**−0.10**	**0.00**
Light level low	**−0.60**	**−0.73**	**−0.46**
log_{10} body mass: light level low	**0.09**	**0.04**	**0.14**
Pace of ecology ballistic predator	**0.14**	**0.04**	**0.25**
Pursuit predator	0.30	−0.01	0.61
Habitat nektonic	0.05	−0.05	0.14
Method ERG	0.04	−0.08	0.15
Random effects			
Phylogeny	0.21	0.12	0.36
Within species	0.01	0.00	0.02
Residuals	0.01	0.01	0.02

Significant values, highlighted in bold, are deemed to be those with 95% of the posterior estimate above or below zero. The model has 137 CFF measures for 121 species. The modes (β) and 95% CI of the posterior distributions are given for all fixed and random terms in a model with log_{10} of CFF as the response variable. Fixed terms include the continuous variable body mass (g) and the categorical terms light level (high, low), pace of ecology (sessile-food foraging, ballistic predator, pursuit predator), habitat (nektonic, demersal) and CFF measurement type (behavioural, ERG). The random terms associated with phylogenetic relatedness (phylogeny), intraspecific variation (within species) and residual variation (residual) are also presented. The h^2 for this model is 0.93 with a lower and upper 95% CI of 0.83 and 0.98.

Mollusca, Echinodermata and Chordata, we find that volant species and species associated with pursuit predation have the highest temporal processing abilities. We also unexpectedly find that several of these patterns are mediated by environment type, with aquatic environments showing associations with higher temporal perception with ballistic predation strategies and in species with smaller body sizes in high light environments. We additionally provide further support for the positive relationships between the tempo of perception and high light environments[11,12] and the use of ERG methods to measure CFF[17,18].

As expected, given the highly dynamic nature of their lifestyle, we found the highest temporal perception abilities in volant species when compared with non-volant species. Although not previously tested within a phylogenetic comparative framework, this link is generally well supported by the frequency of high CFF values reported in volant species. For example, the highest CFF values found across the animal kingdom are typically found in volant groups, particularly those with ecologies that require high manoeuvrability such as dragonflies and tsetse flies[19,20]. This pattern is also seen within vertebrates, with the highest CFF values associated with bird species such as pied fly catchers and raptors[14,15]. The importance of volancy in driving higher temporal perception abilities is also highlighted by the consistently high CFF values found in the volant species in our dataset, with species in low light environments (such as several moth species) having higher CFF values compared with non-volant species in brighter conditions. The link between higher CFF values and highly dynamic ecologies is also supported within volant taxa. For example, Potier et al.[15] found that among three raptor species, the highest CFF values were associated with the species with the most rapid movements and that hunts faster moving prey.

Similar to volancy, we show that the link between trophic ecologies associated with capturing rapid prey and a higher perception tempo is a general pattern found across taxonomic groups and environments. The importance of this link has previously been highlighted and supported by studies comparing visual temporal processing within taxonomic groups such as deep-sea crustaceans. More generally, it is notable that the lowest perception tempos of 4 Hz and below are typically found in species with sessile food sources, such as the crown-of-thorns starfish[10], the giant African land snail[21] and deep-sea scavenging isopods[16]. Mechanistic evidence of the importance of predation in driving the tempo of perception has also been demonstrated in swordfish, which have adaptations to increase their temporal resolution while hunting[22]. However, although we find clear evidence of increased CFF values in species linked with pursuit predation strategies in our main model, the link is less clear in species that rely more on ambush strategies to capture prey.

In contrast to pursuit predators, we find that the link between the tempo of perception and species with less protracted predation strategies, such as ambush predators, is mediated by the type of environment. Although we find strong support for a positive effect associated with ballistic predation strategies in aquatic environments, we do not find support for this relationship for species in terrestrial environments. This context dependency may be due to the high-density nature of liquids, which enable relatively instantaneous changes in motion, thus better facilitating ballistic predation strategies, in comparison to non-volant terrestrial species that are more restricted in their ability to instantaneously change direction[23]. For example, terrestrial species that use ballistic appendages, such as tongues in plethodontid salamanders and in chameleons, or those which lunge over considerable distances, such as jumping spiders, may track and target prey, but once the interaction is initiated, the trajectory has relatively low levels of manoeuvrability[24,25]. These species may in turn rely on strategies that approximate future prey position over the fine-scale continuous updating of prey position. Such a case is demonstrated in tiger beetles, which are observed to use intermittent movement patterns when hunting prey, as during motion they exceed the temporal processing abilities of their visual system, which requires them to periodically stop to recalibrate prey position[26]. This potential difference in the ability to act on fine-scale information in aquatic environments may also explain the stronger negative association between body mass and CFF found in aquatic environments, which supports previous findings that smaller, more manoeuvrable species have a higher tempo of perception[17]. Conversely, integrating sensory signals over longer time windows may be more beneficial in terrestrial conditions where the sensory horizon is much longer, favouring more proactive, rather than merely reactive, behavioural control strategies[27]. Surprisingly, we found the reverse relationship between body size and CFF for species in low light aquatic environments, with larger species associated with higher CFF values. This relationship may highlight how larger eyes have the capacity to capture more photons and may be able to offset the need for longer integration times in low light environments. However, such an offset of this trade between integration time and temporal perception may only be possible in larger species, such as swordfish, which can evolve eyes with the absolute size required to capture enough light, potentially leading to the difference in body size relationship found here. Our results suggest that the tempo of perception may be more strongly linked to particular feeding strategies within certain environmental contexts, with further exploration of temporal perception abilities across different trophic strategies likely to be a fruitful future approach to further our understanding of the evolution of temporal processing abilities.

Our phylogenetic analysis finds high h^2 values across all models, indicating that CFF values are more similar for closely related species and suggesting that CFF is a relatively evolutionarily conserved trait. These values reflect the macroecological scale of our analysis and how CFF can be conserved within certain taxonomic groups. For example, although there is variation associated with light levels in Lepidoptera[8], relatively similar and consistently high CFF values are observed across this taxonomic group. However, although CFF may be somewhat phylogenetically conserved at macroecological scales,

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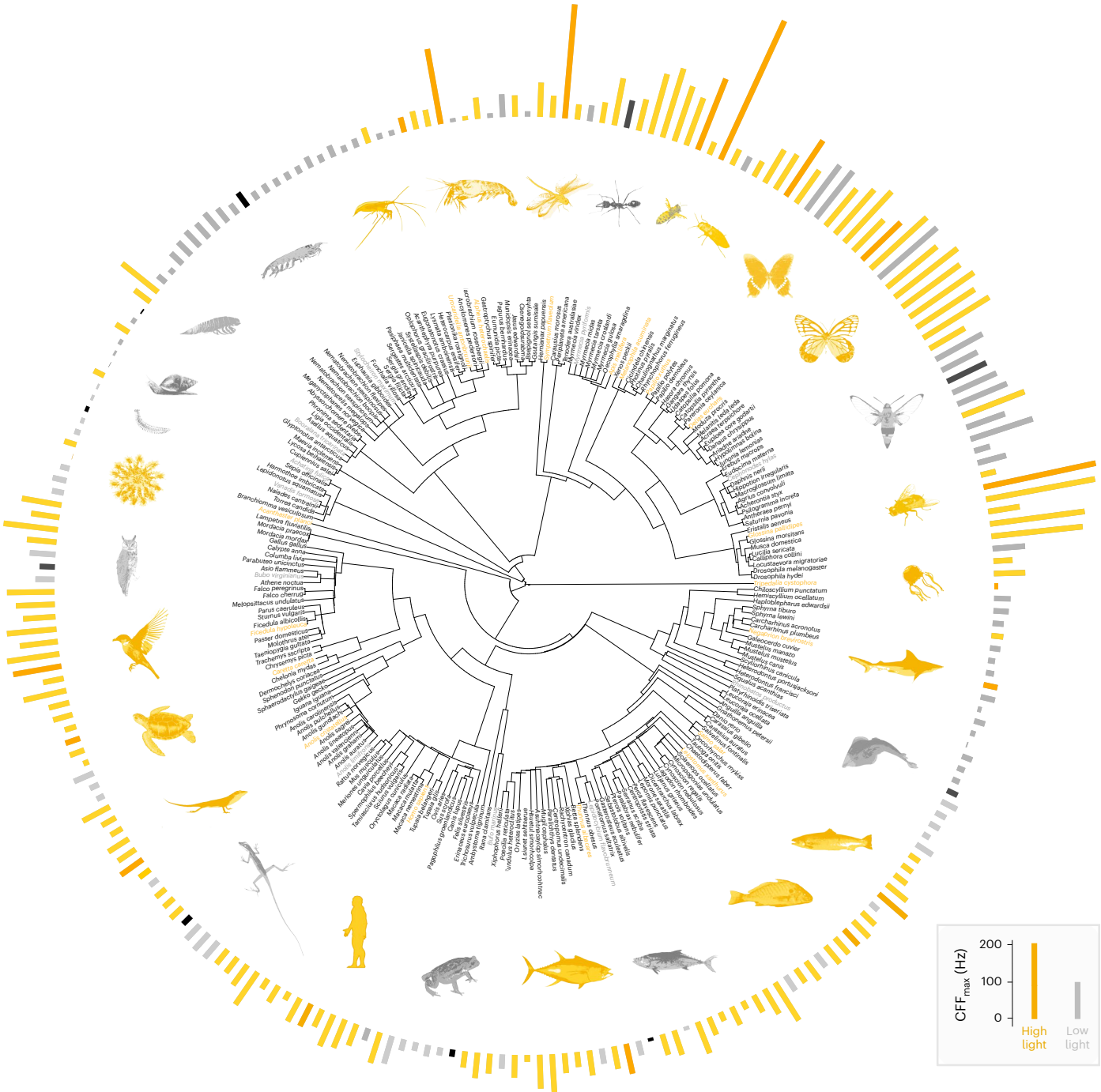


Fig. 2 | Phylogenetic relationship of 208 species included in the analysis. The maximum CFF value associated with each species is represented by bars, with grey bars representing species in low light environments ($n = 107$) and yellow bars representing species in high light environments ($n = 130$). A select range of species are highlighted with silhouette with their corresponding species names and CFF bars are shown in a darker shade.

it is clearly labile under evolutionary pressure, as demonstrated in the isopods *Booralana tricarinata*, a deep sea scavenger and its relative *Ligia occidentalis*. The latter being an intertidal species, it experiences more variation in luminance levels and is predated upon by both terrestrial and marine predators. CFF between these two species differs with over 100 Hz, probably reflecting the different evolutionary pressures on these species. Wide variation in CFF has also been observed between other closely related species, such as in sharks[28]. Along with the variation within phyla shown in Fig. 2, this illustrates that CFF is probably a very flexible trait that can be evolutionarily tuned up or down across species depending on ecological utility and offsetting costs. Although our analysis focused on macroecological scales, and hence mainly variation among species, CFF also varies among individuals within a species, as observed in humans[29], indicating that it is probably linked to genetic variation, which provides a selectable substrate for natural selection. The underlying biological mechanisms that are affected by genetic variation are unknown but could involve alterations in the baseline setpoints of various physiological parameters of cells or connections in the retina[30–32] or parameters affecting the tempo of oscillations in higher visual processing systems in the brain[33,34]. Thus, the evolutionary costs and limitations to high CFF would probably be the considerable amount of energy that the rapid neural processing of visual information demands, as well as a need for an adequate physiological framework to carry out this processing. This need not only

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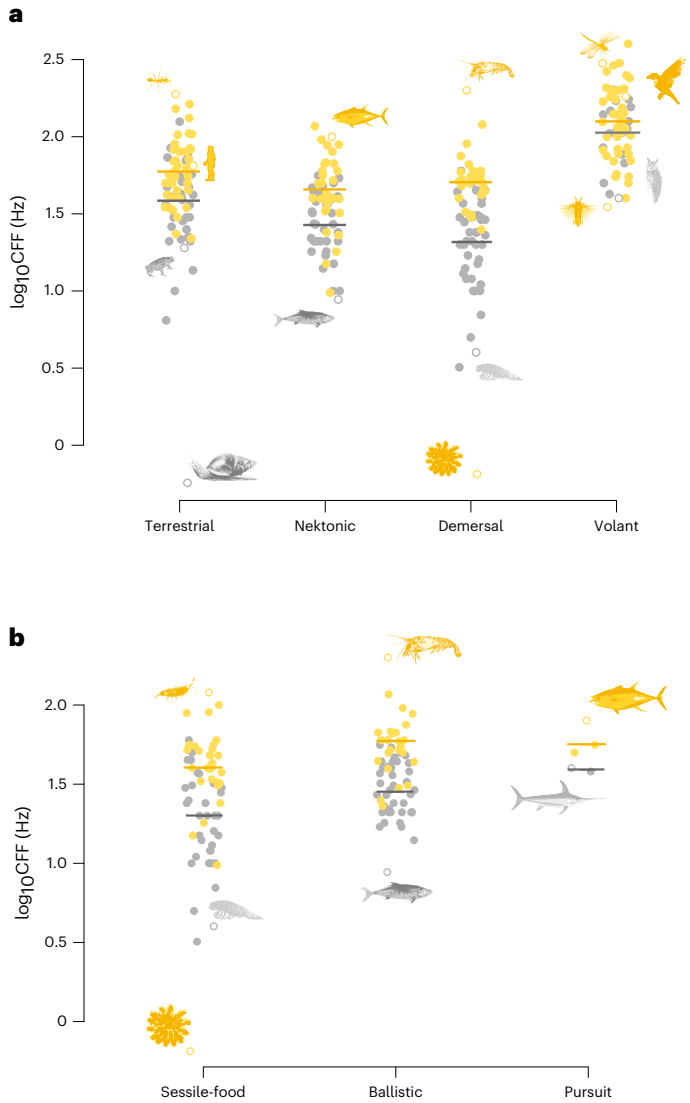


Fig. 3 | The $\log_{10}$ of CFF (Hz) for each of the main habitat types and for each of the foraging modes of life for aquatic species. a, Yellow points represent species in high light environments ($n = 130$), and grey points represent species in low light environments ($n = 107$). Lines represent median values for each of the locomotive groups, including terrestrial species ($n = 55$), nektonic ($n = 57$), demersal ($n = 65$) and volant ($n = 61$). Hollow points represent silhouette species, which are from left to right; *Bufo marina*, *Myrmecia croslandi*, *Achatina fulica*, *Homo sapiens*, *Lepidocybium flavobrunneum*, *Thunnus albacares*, *Acanthaster planci*, *Booralana tricarinata*, *Alpheus heterochaelis*, *Xenos peckii*, *Sympetrum flaveolum*, *Bubo virginianus*, *Ficedula albicollis*. **b**, Yellow points represent species in high light environments ($n = 54$), and grey points represent species in low light environments ($n = 67$). Lines represent median values for each of the foraging modes including sessile-food foraging species ($n = 61$), ballistic predators ($n = 57$) and pursuit predators ($n = 4$). Hollow points represent silhouette species, which are from left to right; *Acanthaster planci*, *Ligia occidentalis*, *Booralana tricarinata*, *Lepidocybium flavobrunneum*, *Alpheus heterochaelis*, *Xiphias gladius*, *Thunnus albacares*.

would be limited to the visual system but would probably apply to multiple aspects of species' physiology, as a high temporal resolution is only useful if the organism has the physical capacity to respond to fleeting information at a suitable pace. Hence, although CFF may be somewhat labile under certain ecological conditions, as demonstrated in isopods, these energetic costs are likely to somewhat constrain temporal perception levels unless there are considerable selection pressures to increase them, as potentially seen in Lepidoptera.

Although our results support Autrum's hypothesis at macroecological scales, they also highlight how the impacts of anthropogenic light pollution may specifically affect species with fast-paced ecologies or higher trophic positions. For example, species with fast-paced ecologies, such as volant species, are predicted to be particularly susceptible to flickering anthropogenic lighting, a potential issue well established in birds[35] but less understood in invertebrates. Similarly, species with pursuit predation strategies, in particular, aquatic predators, are also expected to be more susceptible to potentially detrimental effects from light pollution. Such a case is demonstrated in bluegill fish (*Lepomis macrochirus*), which show reduced prey capture and strike rates under flashing high-intensity lights[36]. However, whether such potential susceptibility for species with fast-paced ecologies to flashing artificial lighting has consistent ecological effects, such as top-down trophic cascades, is currently poorly understood[37], highlighting the need to further understand temporal perception within an ecological framework.

Methods

Data

To test our hypothesis that temporal perception relates to species ecology, we collated data on maximum CFF, body mass, habitat light conditions, foraging strategy and the general habitat niche of a species. For CFF, we first used the values from Lafitte et al.[38], which include maximum CFF values reported under experimental conditions for 156 species. Second, we searched literature up to 1 September 2024 using Web of Science for additional CFF values, using the same search terms mentioned in Lafitte et al. Third, we screened all collected publications for additional references containing CFF values. In our analysis, we only included species in this database that used either ERG or behavioural experiments, which include optomotor responses. We did not include species in the dataset which were only tested under a restricted set of wavelengths. For example, we did not include the study of *Trigona spinipes*, which only tested responses on green photoreceptors[39]. All publications were screened by the authors C.Ha. and K.H., and several publications were additionally screened by C.Hy. See Supplementary Sections 2 and 3 for full literature search details, including the PRISMA workflow and exclusion criteria.

For body size, we collated measures of mean adult body mass (g) from the amniote database for mammals, birds and lizards[40], from FishBase for fish species[41], SeaLifeBase for marine invertebrates when available[42] and various field guides for the remaining species (Supplementary Section 4). For four polychaete species that did not have available body mass data, we estimated body mass using conversion methods based on converting volume estimates from photographs[43]. Further details on the estimates of body mass are outlined in the Supplementary Section 4.

For habitat type and foraging mode, we used descriptions from various sources, including Mammals of the World[44], Handbook of Birds of the World[45], FishBase[41] and SeaLifeBase[42]. For habitat, we defined species in non-aquatic environments as either terrestrial or volant (flying) and species in aquatic environments as nektonic (free-swimming) or demersal (living on or near the bottom). Only species with powered flight were defined as volant, with species such as *Gallus gallus* defined as terrestrial. We defined aquatic species as nektonic or demersal on the basis of descriptions in Fishbase[41] and SealifeBase[42] and on additional descriptions in literature when available. Habitat light conditions were defined on the basis of the light levels of typical foraging conditions, with diurnal species that forage in open habitats or shallow waters designated as having high light conditions, whereas species described as nocturnal or that foraged in deeper waters were designated as having low light conditions.

To define foraging type, we used the three general categories 'sessile-food foraging', 'ballistic predator' and 'pursuit predator'. Herbivores, scavengers and species that feed on items that are effectively sessile and require no element of pursuit were defined as sessile-food

foraging species. Species that are described as pursuing prey over a sustained period of time extending past the initiation of the trophic interaction were defined as pursuit predators. Finally, species that are described as 'sit and wait' or 'ambush' predators or are not described as engaging in any pursuit post the initial phase of the trophic interactions were defined as ballistic predators.

To correct for non-independence due to common ancestry, and to quantify the phylogenetic signal within our analysis, we built a chronogram (Fig. 2) using available phylogenies and time-calibration points for major ancestral branching, following the recommendations of Parham et al.[46]. For Aves, Mammalia, squamates, turtles, amphibians, bony fish and elasmobranchs, we used the phylogenies from Jetz et al.[47], Kuhn et al.[48], Pyron et al.[49], Thomson et al.[50], Pyron and Wiens[51], Betancur et al.[52] and Stein et al.[53]. For the remaining species, we pruned the metazoan phylogeny from the Open Tree of Life[54], attached each of the dated phylogenies to the pruned Open Tree of Life phylogeny and dated major ancestral nodes using the range of time-calibrated estimates obtained from several sources (see Supplementary Section 5 for details on each estimated ancestral date). As uncertainty in both topology and branch lengths of a phylogenetic tree cannot be captured in a single phylogeny, we accounted for this uncertainty by using a distribution of 100 phylogenetic trees in our phylogenetic analyses. We generated this distribution by sampling trees without replacement from the available distributions of phylogenies for the phylogenies from Jetz et al.[47], Kuhn et al.[48], Thomson et al.[50] and Stein et al.[53]. To account for the uncertainty associated with the date of each ancestral node, each tree was constructed by drawing node dates from a uniform distribution on the basis of the reported uncertainty range for each ancestral node (see Supplementary Section 5 for more details).

Analysis

To test our predictions, we fitted Bayesian multivariate phylogenetic mixed models using the MCMCglmm package[55] in R version 4.4.1[56]. We fit the main model with $\log_{10}$ CFF as the response variable and $\log_{10}$ body mass (g), light level (high, low), pace of ecology (sessile-food foraging, ballistic predator, pursuit predator), habitat (terrestrial, nektonic, demersal, volant) and CFF measurement type (behavioural, ERG) as the explanatory variables. We also included an interaction term between mass and light levels to test for different relationships between these environments. This gave us the general model;

$$\begin{aligned}\mathrm{CFF} = f(&\log_{10}\mathrm{mass} + \mathrm{light\ level} + \mathrm{pace\ of\ ecology} + \mathrm{habitat}\\ &+\mathrm{measurement\ type} + \log_{10}\mathrm{mass} : \mathrm{habitat}).\end{aligned} \quad (1)$$

To test these relationships within the context of just aquatic environments and non-aquatic environments, we ran two separate submodels: one which only included non-aquatic species and another with just aquatic species. As the CFF associated with the black fire beetle (*M. acuminata*) may primarily relate to thermal sensory functions and not functions associated with pace of ecology[57], we also ran a supplementary analysis that removed this species (Supplementary Section 3).

Non-independence of data due to common descent were controlled for by including the phylogenetic relationship between species using the animal term in the MCMCglmm model[55]. The relative variance attributable to the phylogenetic random effect component (h^2) was calculated as the ratio of variance explained by phylogeny to the sum of phylogenetic variance, species variance and residual variance[58]. To account for uncertainty in the phylogeny, we ran our models across the full distribution of 100 trees using the multree package[59] in R and present the combined posterior distributions across this distribution of phylogenies. For species with multiple measures, such as those with both behavioural and ERG measures, variation was accounted for using a separate species-level random term. All models were fitted with parameter expanded priors, and the burn-in, thinning and number of iterations were determined for each model separately to ensure effective sample sizes exceeded 1,000 for all parameter estimates. We ran three separate chains in all models and tested convergence using the Gelman–Rubin statistic[60]. The significance of each variable's effect on CFF was determined on the basis of whether the 95% credibility interval of the variable's posterior distribution was either below or above zero.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The CFF dataset and literature review details are available via figshare at https://doi.org/10.6084/m9.figshare.30556475 (ref. 61).

Code availability

The analysis code is available via figshare at https://doi.org/10.6084/m9.figshare.30556475 (ref. 61).

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Author contributions

C.S.H., C.H. and K.H. conceived the research. C.S.H. and C.H. collected the data. C.S.H., C.H. and K.H. carried out the statistical analyses. K.H., A.L.J., K.J.M. and R.G.O. supervised the project. C.S.H., C.H. and K.H. wrote the paper with contributions from A.L.J., K.J.M. and R.G.O.

Competing interests

The authors declare no competing interests.

Additional information

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection
Data analysis	Analyses were carried out with R version 4.4.1. The following packaged were used: - "MCMCglmm" for Bayesian analyses - "phytools," "caper" and "multree" for phylogenetic analyses - "hdrcde" for density estimation

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Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

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Population characteristics	This research did not involve human participants
Recruitment	This research did not involve human participants
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Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study examined links between the pace of visual perception and species' ecology. We used species critical flicker fusion thresholds (CFF), a measure for visual temporal resolution, as our response variable. We used the following ecological measures as explanatory variables: body mass, habitat light conditions, foraging strategy, habitat niche and CFF measuring type. We included and interaction term between body mass and habitat type.
Research sample	The research sample consists of 237 species, as collected from available previous literature. Some exclusion criteria were employed, which are reported in the dataset. Sample is meant to represent general trends and relations among the tested variables within the animal kingdom.
Sampling strategy	Starting point for sampling was a previous publication which had collated data from literature (Lafitte et al., 2022. See manuscript reference list and Method section). We then also we also added additional studies using forward and reverse citation searches using Web of Science. Values for explanatory variables were collected in online and offline archives, catalogues and literature as outlined in the Supplementary material.
Data collection	Data was collected through literature search, using Web of Science, Google Scholar and Google. Data collection was carried out by Clinton S. Haarlem, Cliodhna Hynes and Keavin Healy.
Timing and spatial scale	Data collection commenced in August 2021 and ended in September 2024. Collection took place periodically and no specific scheduling was used to conduct searches.
Data exclusions	Data that did not report specific CFF values were excluded, as were studies that measured only a specific wavelength (colour) and studies that did not measure intact visual systems.
Reproducibility	Experimental findings were in agreement with previous literature. Due to the nature of the study, we only analysed data available from previous studies and did not produce original data. As such, no specific measures for reproducibility were put in place.
Randomization	Data was allocated into groups based on specific variable factors used in analyses, such as habitat type and foraging strategy. Data was allocated for each variable factor as described in the Methods section.
Blinding	No blinding was used in this study. However, data was collected by three researchers independently and each sample provided by one researcher was assessed for suitability by the other two.

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Materials & experimental systems

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☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	No plants were used
Novel plant genotypes	No plants were used
Authentication	No plants were used